M3 Baseline Simulator and Verification Evidence

Differentiable Self Calibration for Fourier Ptychography

Team 25 | Abhishek Jain, Aarya Anup, Saeed Javadizadeh, Raymond Yao

Repository. https://github.gatech.edu/[OWNER]/project-team-25 | Commit to grade: [ADD SHORT COMMIT]

Changes since M2. The M3 cold run keeps the 7 x 7 illumination set but reduces the object from 256 x 256 to 64 x 64, uses one fixed synthetic seed, and omits real-data and multi-seed claims. This is the smallest version that runs quickly while testing the same fixed-versus-joint calibration question.

Team Contributions

Member	Contribution statement to confirm before submission
Abhishek Jain	[Add verified contribution.]
Aarya Anup	[Add verified contribution.]
Saeed Javadizadeh	[Add verified contribution.]
Raymond Yao	[Add verified contribution.]

1 How to Run It

Environment. Run from the repository root. No download, account, absolute path, or external dataset is required.

Command	Observed time	Expected result
conda activate cse6730	< 1 s	Activates the course environment.
python -m pytest -q	about 1 s	Last line: 3 passed; writes verification JSON records.
python scripts/make_figures.py	about 17 s	Writes results/m3_first_result.png and results/m3_summary.json.

Fresh-clone check. The packaged ZIP was unpacked and rerun on October 22, 2026: pytest reported 3 passed in 0.31 s and the figure rebuilt in 17.1 s. Independent teammate confirmation: [add name/date].

2 Verification Evidence

Check and failure rule. On a 32 x 32, 3 x 3 LED, noise-free case, compare analytic loss gradients for dx, dy, and theta with centered finite differences. The check fails if the maximum relative error at h = 10^-4 exceeds 10^-4 or does not decrease as h is refined.

Finite-difference step h	Maximum relative error	Result
10^-2	1.329 x 10^-3	Refine
10^-3	1.329 x 10^-5	Pass
10^-4	1.329 x 10^-7	Pass

Evidence and scope. tests/test_gradient.py runs this check under pytest and records results/gradient_check.json. The error falls by about 100x per tenfold refinement and the finest result is 752 times below the threshold. This supports the implemented pose derivatives; it does not validate the optical assumptions, optimizer convergence, or experimental accuracy.

3 Baseline and First Result

Baseline. The NumPy baseline models a coherent thin object, circular known pupil, 49 intensity frames, and a rigid LED pose (dx, dy, theta). Both arms use the same amplitude-loss optimizer; the fixed arm holds pose at zero and the joint arm estimates pose after 25 warm-up iterations. Pupil aberration, per-LED errors, real data, grid refinement, and repeated seeds are deferred to M4.

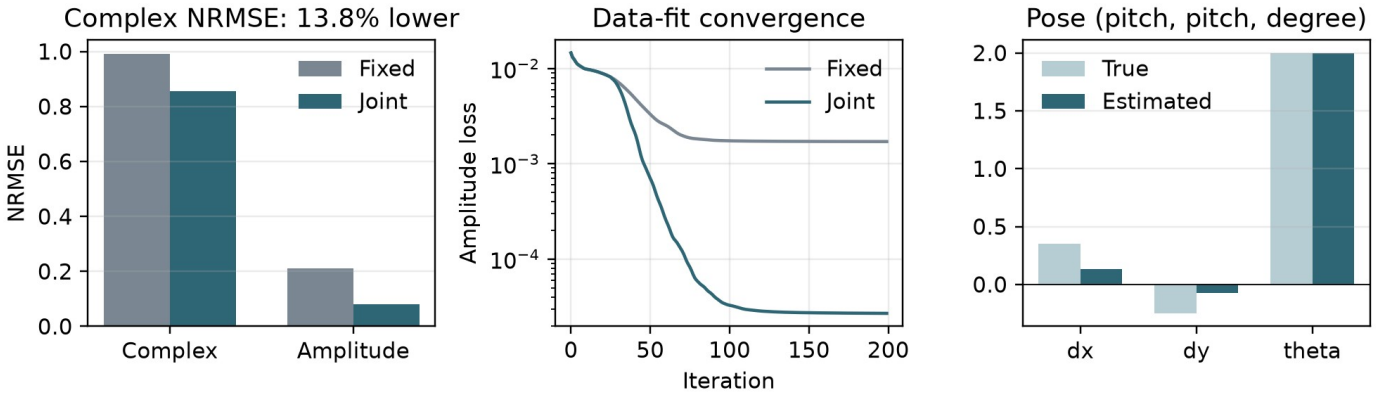


Figure 1. Fixed nominal geometry versus joint pose estimation for seed 20261022.

Quantity	Fixed	Joint	Change
Aligned complex-object NRMSE	0.9925	0.8553	13.8% lower
Amplitude NRMSE	0.2099	0.0805	61.6% lower
Final amplitude loss	1.708 x 10^-3	2.694 x 10^-5	98.4% lower

Interpretation. For this one reduced case, joint calibration lowers the planned complex-object QoI and recovers rotation (1.999 degrees versus 2.000 degrees true). It does not yet establish the M1 claim: translation remains biased, absolute complex NRMSE is high, and ten matched seeds plus grid refinement are still required.

4 Technical Risks

Risk	How we will notice it	Response
Global translation is confounded with an object phase ramp.	dx or dy stays biased while loss falls and rotation converges; repeat across the 10 M2 seeds by Oct 30.	Fix an object-phase gauge or add a calibrated target/phase prior; report pose only after a known-answer identifiability test.
The nonconvex optimizer depends on initialization.	Fixed and joint NRMSE or pose estimates vary materially over 10 matched starts.	Use matched multi-start runs, retain the best verified loss per arm, and report medians with bootstrap intervals.
Unknown pupil aberration is absorbed as LED pose.	Recovered pose changes or residuals remain structured when low-order pupil terms are enabled.	Compare ideal, supplied, and jointly estimated pupils; add only identifiable Zernike terms to the M4 model.

5 Graduate Extension Check

What ran. scripts/extension_prototype.py evaluates analytic derivatives of one defocus-like and one astigmatism-like pupil coefficient and compares them with centered finite differences. It runs in about 0.05 s; pytest also executes it.

Evidence. The maximum relative derivative error was 1.03 x 10^-8, below the 10^-4 M2 threshold; results/extension_gradient_check.json records both gradients. This verifies the two-parameter pupil phase derivative only, not joint recovery from intensity data.

Next step. Insert both coefficients into the imaging loss, then recover nonzero synthetic coefficients within 10% while keeping pose degradation below 0.02 LED pitch and 0.02 degree. This depends on first resolving the translation-phase gauge and stabilizing the pose-only baseline.

Before Submission

Required final entries. Replace the four contribution placeholders, repository owner, short commit, and fresh-clone tester/date. Push that default-branch commit, add the five course collaborators, and complete the individual Canvas peer evaluation.